

pedk.ai - AI-native operational reconciliation engine

Market Research Outcome — Version 1.0

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1 pedk.ai — Product Overview

AI-Native Operational Reconciliation Engine for Financial Market Infrastructure

1.1 1 What pedk.ai** delivers**:

- Complete operational intelligence platform (alarms, incidents, topology, telemetry, capacity, reports)
- TMF 642/628 API compliance (customers increasingly require standards compliance)
- Multi-domain telemetry ingestion and analysis
- CMDB management and enrichment
- Alarm correlation and noise reduction
- Sleeping cell and divergence detection
- Local deployment (data sovereignty for operators in regulated markets)

1.1.1 1.1 Summary: Mechanism-by-Mechanism Competitive Matrix

Mechanism	Nokia	ServiceNow	NetoAI	Google	Huawei	pedk.ai
Latent evidence retention	No	No	No	No	No	Yes (unique)
Multi-modal embedding fusion	No (text-based vector store)	No (statistical correlation)	T-VEC (semantic only)	No (graph queries)	No (dynamic baselines)	4-column (sem+topo+temp+ops)

Mechanism	Nokia	ServiceNow	NetoAI	Google	Huawei	pedk.ai
Topology-aware correlation	Ontology engine	CMDB-based grouping	DigiTwin knowledge graph	Temporal graph (Spanner)	Multimodal AI topology	Shadow Topology + 2-hop enrichment
Closed-loop weight optimization	"Reflection" (vague)	Not disclosed	Not disclosed	Not disclosed	Not disclosed	Bayesian per-profile (documented)
Hypothesis generation	Not disclosed	LEAP (6-month mining)	TSLAM reasoning	Gemini reasoning	Not disclosed	TSLAM-8B falsifiable claims
Causal testing	No	5 correlation algorithms	Not disclosed	Not disclosed	Not disclosed	Causal Direction + Expectation Violation
Local model serving	Cloud-implied	Cloud (Now platform)	Cloud + on-prem option	Cloud (Vertex AI)	Cloud (NAIE)	Local-only (T-VEC+TSLAM)

1.1.2 1.2 Summary: Service Management Competition to pedk.ai

Factor	BMC Helix	ServiceNow
Competitive threat	Low (no latent evidence capability, storage-constrained)	High (Knowledge Graph + agent memory could evolve toward AM)
Gap AM fills	2TB storage cap + no long-horizon historical analysis + no cross-domain semantic fusion	"Persistent memory" in agents is evolving – AM's gap may close

Factor	BMC Helix	ServiceNow
Strategic posture	Splitting from BMC in 2025 – actively seeking differentiation and partnerships	Dominant market position – less incentive to partner with startups
ISV program	Formal ISV program with Marketplace listing, sandbox access, product certification, developer portal	Larger marketplace but harder to get visibility
eTOM/TMF alignment	Native eTOM/SID modeling – AM's fragment source types map naturally	General ITSM first, telecom extensions second
Pricing headroom	Cheaper than ServiceNow – combined solution still undercuts ServiceNow total cost	ServiceNow TCO already 3-5x license cost – adding AM increases sticker shock

1.1.3 1.3 Incumbent Pricing Landscape

Player	Pricing Model	Estimated Cost	Notes
ServiceNow ITSM	Per agent/month	\$100-160/agent (Standard-Pro)	ITOM adds \$150-200/agent. TCO = 3-5x license cost
ServiceNow TSOM	Custom quote + ITSM base	Undisclosed (premium add-on)	Predictive Intelligence is additional
BMC Helix	Named user/month	~\$115/agent	18-21% premium over average. AI features require separate backend token charges
Nokia AVA/AgenticOps	Bundled with network equipment or aaS/hosted/hybrid	Undisclosed	Flexible delivery: aaS, hosted, hybrid

Player	Pricing Model	Estimated Cost	Notes
Ericsson Operations Engine	Managed services model (outcomes-based)	Undisclosed	"Committed business outcomes" pricing. rAppaaS on AWS Marketplace
Huawei ADN	Bundled with equipment	Undisclosed	"Network optimization consumes >30% of operational costs"
Tupl	SaaS monthly subscription	Undisclosed but "no strings attached, option to stop any time"	First to offer SaaS for network operations automation
NetoAI	Unknown (unfunded startup, founded 2024)	Likely usage-based or enterprise license	Small team, no published pricing

Critical insight: A global telecom firm was paying \$40,000/year for 220 standalone ServiceNow Discovery licenses *while already having ITOM subscriptions that included the same entitlement*. This illustrates the pricing opacity and accidental overspend that characterizes the incumbents.

1.1.4 1.4 Market Segmentation by Budget

Segment	Typical IT Ops Budget	Current Solutions	Pain Points	AM Opportunity
Tier-1 (AT&T, Vodafone, DT)	50M – 200M + ops/year ServiceNow + TEM – native + custom internal tools Integration complexity, vendor lock-in, diminishing ROI from incumbents *Add – onlayer * * : AM as intelligence enhancement on top of existing ServiceNow / BMC sta per incident)			

Segment	Typical IT Ops Budget	Current Solutions	Pain Points	AM Opportunity
Tier-2 (MasOrange, TPG, TalkTalk)	\$5M-50M ops/year	Mix of BMC/ServiceNow + TEM-native	Cost pressure, fewer engineering resources, harder to justify ServiceNow total cost	Primary platform play: AM + BMC Helix integration at lower total cost than ServiceNow standalone
Emerging market (Jio, Airtel, African operators)	\$1M-20M ops/year	In-house + limited TEM tools + open source	Extreme cost pressure, massive scale, data sovereignty requirements	Embedded/partnership: AM as the abeyance layer within operator-built stacks. Local model serving is decisive advantage. Usage-based pricing
Smaller/regional operators	<\$1M ops/year	Basic NMS + spreadsheets	Cannot afford ServiceNow/BMC. Need "good enough" at fraction of cost	SaaS: Tupl-style monthly subscription, self-service deployment. AM Lite (Layer 1-2 only)

1.1.5 1.5 What pedk.ai Actually Has Today (Not What It Plans to Build)

The codebase audit reveals a **substantially built product**:

Capability	Implementation Status	Competitive Relevance
120+ API endpoints (alarms, performance, incidents, topology, capacity, policies, reports)	90% functional	Full operational intelligence platform
TMF 642 (Alarms) + TMF 628 (Performance) standard APIs	Implemented	Standards compliance that operators require
ServiceNow integration (poll incidents, correlate actions, behavioral feedback loop)	95% operational	Enhances existing ServiceNow, not replaces it
CMDB sync (Datagerry adapter, entity deduplication, incremental sync)	90% operational	CMDB enrichment without manual effort
6-domain telemetry ingestion (RAN, transport, fixed broadband, core, enterprise, power) via Kafka	100% operational	Multi-domain visibility from a single platform
Telecom-specific embeddings (T-VEC 1536-dim, zero cloud cost)	100% operational	Better semantic search than generic models, locally served
TimescaleDB metrics with KPI storage	100% operational	Time-series analytics on operator data
Shadow Topology (entity resolution, 2-hop BFS, confidence scoring)	Fully coded	Automated network relationship discovery
12 frontend pages (dashboard, incidents, topology, scorecard, sleeping cells, divergence, ROI, feedback, admin, settings)	80% functional	Operator-facing UI ready for demo/pilot
Real-time SSE streaming for alarms	Operational	Live monitoring capability
Snap Engine (5-dimension weighted scoring)	Fully coded	Cross-domain alarm correlation

Capability	Implementation Status	Competitive Relevance
Sleeping Cell detection	Implemented	Identifies idle/degraded cells – immediate operator value
Autonomous Shield (recommendations, not auto-remediation)	Implemented	Safe, human-in-the-loop suggestions
Local LLM inference (Ollama + T-VEC, zero cloud dependency)	Operational	Data sovereignty, zero marginal inference cost
Multi-tenant architecture (tenant isolation at DB layer)	Enforced	Can serve multiple customers from shared infrastructure
Cloud deployment (Oracle Cloud Always Free, Docker, K8s manifests)	Validated	Running infrastructure at near-zero hosting cost

1.1.6 1.6 What Makes pedk.ai work Even Without Unique Differentiation

The competitive moat in this scenario is not technology – it is **cost structure + domain expertise + deployment simplicity**:

1. **Cost advantage is structural, not temporary.** Local model serving (zero cloud LLM cost), Oracle Cloud Always Free hosting, no per-agent licensing, single-engineer efficiency – these are not pricing gimmicks. They reflect a fundamentally lighter cost structure than ServiceNow (thousands of employees, enterprise sales organization, data center costs) or BMC (similar overhead). **pedk.ai** can profitably sell at 50K-200K GBP what ServiceNow charges 500K-2M GBP for.
2. **Telecom domain specificity is a filter.** Of 560+ AIOps companies, the vast majority target generic IT operations. **pedk.ai**'s TMF APIs, telecom-specific embeddings (T-VEC), multi-domain telemetry ingestion (RAN/transport/core/fixed/enterprise/power), and sleeping cell detection immediately filter out generic competitors. The relevant competitive set is perhaps 20-30 companies, not 560.
3. **Data sovereignty is becoming non-negotiable.** EU regulations, Indian DPDP Act, African data localization laws are tightening. **pedk.ai**'s local-only deployment (T-VEC on CPU, Ollama on local GPU, PostgreSQL on-prem) is not a limitation – it is a procurement qualifier in growing markets. Google (Vertex AI), ServiceNow (SaaS), and Nokia (cloud-implied) cannot match this without significant re-architecture.
4. **The consulting-led approach builds the right kind of credibility.** Tupl succeeded

by being founded by telecom/AI domain experts who could walk into a NOC and immediately understand the problem. **pedk.ai**'s founder has this expertise. Consulting engagements prove competence before the product is on trial. One operator saving \$40M (as Tupl's Tier-1 customer did) creates unstoppable word-of-mouth.

5. **Small is an advantage in this market segment.** Operators report frustration with vendor procurement processes that take 6-9 months. A single-person company with a deployed product can offer a 2-week trial, a 1-month pilot, and a month-to-month contract. No 3-year lock-in. No implementation consulting army. This is the Tupl playbook ("no strings attached, option to stop any time") and it resonates with budget-constrained operators.